

Applying Exponent Laws

MA.8.NSO.1.3

I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

Yesterday: a negative exponent means reciprocal.

Today the base is a fraction — so the reciprocal flips it.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: 2^{-3}
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$
3. Which is larger, 4^3 or 3^4 ?
4. Write the reciprocal of $-\frac{2}{5}$

Warm-Up

Answers.

1. Find the value: 2^{-3} $\frac{1}{8}$
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$ 7^2
3. Which is larger, 4^3 or 3^4 ? 3^4 (**81 > 64**)
4. Write the reciprocal of $-\frac{2}{5}$ $-\frac{5}{2}$

Notes

Copy the table.

I. The pattern with a fraction base

Expression	Expanded	Value
$(2/5)^2$	$2/5 \cdot 2/5$	$4/25$
$(2/5)^1$	$2/5$	$2/5$
$(2/5)^0$	1	1
$(2/5)^{-1}$	$5/2$	$5/2$
$(2/5)^{-2}$	$5/2 \cdot 5/2$	$25/4$

Notes

II. Flip the base, make the exponent positive

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$$

$$\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4} \quad \left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

The sign stays with the fraction. Flip the numbers, not the sign.

A whole number is a fraction over 1:

$$4^{-3} = \left(\frac{4}{1}\right)^{-3} = \left(\frac{1}{4}\right)^3$$

Notes

III. Several laws at once — the plan is still yours

$$\left(\frac{2.5^5}{2.5^3}\right)^{-1} = (2.5^2)^{-1} = 2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

Any correct order of laws gives the same value. Finish with a positive exponent.

Example 1

Flip, then evaluate.

$$\left(-\frac{7}{8}\right)^{-2}$$

Rewrite with a positive exponent, then find the value.

Example 1

Worked.

$$\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2 \quad \text{flip the base; the sign stays}$$

$$\left(-\frac{8}{7}\right)^2 = \frac{64}{49} \quad \text{even power: positive}$$

Answer: 64/49

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{5}{2}\right)^{-3}$$

Your Turn

Answer.

$$\left(-\frac{5}{2}\right)^{-3}$$

$(-2/5)^3$ — odd power, negative

Answer: $-\frac{8}{125}$

Example 2

Two laws, then a flip.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

Write an equivalent expression with one base and a positive exponent.

Example 2

Worked.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4 = \left(\frac{5}{8}\right)^{-2+4} = \left(\frac{5}{8}\right)^2 \quad \text{product of powers}$$

$$\left(\frac{5}{8}\right)^2 = \frac{25}{64} \quad \text{already positive: no flip}$$

Answer: $(5/8)^2 = 25/64$

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

Your Turn

Answer.

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

exponent -10 , then flip: $(-4/5)^{10}$

$$\textbf{Answer: } \left(-\frac{4}{5}\right)^{10}$$

Whiteboards • Question 1 of 9

Boards up on three.

A copier is set to shrink a page to $\frac{2}{3}$ of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

What fraction of the original page's size is the last copy?

Answer it.

Whiteboards • Question 1 of 9

Answer.

A copier is set to shrink a page to $\frac{2}{3}$ of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

What fraction of the original page's size is the last copy?

$$\left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4}$$

Answer: $\frac{16}{81}$

Whiteboards • Question 2 of 9

Boards up on three.

$$\left(\frac{3}{-5}\right)^{-9}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 2 of 9

Answer.

$$\left(\frac{3}{-5}\right)^{-9}$$

flip the numbers; the sign stays in the fraction

Answer: $\left(\frac{-5}{3}\right)^9$

Whiteboards • Question 3 of 9

Boards up on three.

$$\left(-\frac{8}{9}\right)^{-2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\left(-\frac{8}{9}\right)^{-2}$$

$$(-9/8)^2 = 81/64$$

$$\text{Answer: } \frac{81}{64}$$

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{21}{4}\right)^{-5}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{21}{4}\right)^{-5}$$

flip

Answer: $\left(\frac{4}{21}\right)^5$

Whiteboards • Question 5 of 9

Boards up on three.

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

A. $(7/8)^3$

B. $(-8/7)^3$

C. $(-7/8)^3$

D. $(8/7)^3$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

A. $(7/8)^3$

B. $(-8/7)^3$

C. $(-7/8)^3$

D. $(8/7)^3$

Answer: B — $(-8/7)^3$

Whiteboards • Question 6 of 9

Boards up on three.

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(8/5)^2$

B. $(5/8)^{-6}$

C. $(5/8)^2$

D. $(5/8)^8$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(8/5)^2$

B. $(5/8)^{-6}$

C. $(5/8)^2$

D. $(5/8)^8$

Answer: C — $(5/8)^2$

Whiteboards • Question 7 of 9

Boards up on three.

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

$$(5/2)^1 \cdot (5/2)^1 = (5/2)^2$$

$$\text{Answer: } \frac{25}{4}$$

Whiteboards • Question 8 of 9

Boards up on three.

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

Answer it.

Write it with one base and one positive exponent.

Whiteboards • Question 8 of 9

Answer.

$$\left[\left(\frac{2}{3} \right)^5 \cdot \left(\frac{2}{3} \right)^{-2} \right]^3$$

$(2/3)^3$ inside, then power of a power

Answer: $\left(\frac{2}{3} \right)^9$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

Write your answer in sentences.

Say what the negative exponent does, and what it does not do. Give the correct rewrite.

Whiteboards • Question 9 of 9

Answer.

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

the exponent flips the fraction; it never touches the sign

Answer: No — $(-2/3)^{-5} = (-3/2)^5$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Identify equivalent expressions involving exponents I — VLM